

A Transport-Closure Foundation for Digital Root Power Tables over $\mathbb{Z}/d\mathbb{Z}$

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Abstract

A Digital Root Power Table, or DRPT, is the finite power table

$$T_d(x, k) = x^k \pmod{d},$$

usually displayed in base 10 with $d = 9$ and the residue $0 \pmod{9}$ shown as 9. This paper treats the DRPT not as a visual pattern table, but as a finite transport object over the ring $\mathbb{Z}/d\mathbb{Z}$.

The central result is a transport-closure theorem: rows, columns, addition, multiplication, and cross-modulus projection all close through a forced coordinate stack consisting of gcd-depth, quotient modulus, quotient-unit position, anchored CRT position, and odd-component mirror refinement. Rows flow through gcd-depth into quotient-unit cycles. Columns act as exponent operators. Addition walks anchored CRT position. Multiplication folds the tile through saturated gcd-depth, then cycles inside the surviving quotient.

The paper also proves a coordinate-necessity theorem. Coarser profiles such as gcd-depth alone, quotient modulus alone, mirror signature alone, or unanchored quotient-unit data lose deterministic transport information. Explicit counterexamples show that deleting these coordinates causes addition or multiplication to become nondeterministic on the induced profile classes. Thus the coordinate system is not introduced by taste or convenience. It is the coordinate stack left standing when deterministic transport closure is demanded.

The claims are intentionally narrow. The paper establishes a finite algebraic substrate of transport-coordinate necessity for the table T_d .

Addition walks the tile; multiplication folds the tile, then spins inside the surviving quotient.

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1 Introduction

Digital root patterns are often treated as base-dependent curiosities. In base 10, one commonly studies the repeating behavior of powers under digital root reduction, which is equivalent to working modulo 9 and displaying the residue $0 \pmod{9}$ as 9.

This paper generalizes that object. For any integer $d \geq 2$, define the finite power table

$$T_d(x, k) = x^k \pmod{d}.$$

The base-10 digital root table is the special display convention

$$d = 9, \quad 0 \pmod{9} \leftrightarrow 9.$$

The term *Digital Root Power Table* is retained for historical continuity with the companion paper [8], but mathematically the object is simply a finite power table over $\mathbb{Z}/d\mathbb{Z}$.

The goal is not to claim that such tables immediately prove number-theoretic results. The goal is more basic: to determine what coordinates are forced if one demands that the basic transport operations of the table close deterministically.

Those operations are

$$x \mapsto x^k, \quad x \mapsto x + y, \quad x \mapsto xy, \quad \mathbb{Z}/D\mathbb{Z} \rightarrow \mathbb{Z}/d\mathbb{Z} \quad (d \mid D).$$

The paper proves that the natural finite-ring coordinates are not arbitrary bookkeeping. They arise because weaker coordinate systems fail to determine transport.

2 Basic definitions

Let

$$d = \prod_{p \mid d} p^{r_p}.$$

For $x \in \mathbb{Z}/d\mathbb{Z}$, choose any integer representative and define

$$g_d(x) = \gcd(x, d).$$

When $x \equiv 0 \pmod{d}$, set $g_d(0) = d$. Define the quotient modulus

$$q_d(x) = \frac{d}{g_d(x)}.$$

If $x \neq 0$, define the reduced quotient-unit coordinate

$$u_d(x) = \frac{x}{g_d(x)} \pmod{q_d(x)}.$$

Then $\gcd(u_d(x), q_d(x)) = 1$. For $x = 0$, the quotient is $q_d(0) = 1$, and the reduced unit coordinate is terminal.

Every residue therefore decomposes as $x = g_d(x) u_d(x)$ inside the quotient $\mathbb{Z}/q_d(x)\mathbb{Z}$. This gives the basic DRPT residue coordinate

$$K_d(x) = (g_d(x), q_d(x), u_d(x)).$$

The row behavior of the power table is governed not only by the initial residue coordinate, but by the whole kernel path

$$K_d(x; k) = \left(g_d(x^k), \frac{d}{g_d(x^k)}, \frac{x^k}{g_d(x^k)} \bmod \frac{d}{g_d(x^k)} \right).$$

The initial coordinate locates the residue. The path coordinate describes its row evolution.

3 CRT position

Let $d = \prod_i p_i^{r_i}$. The Chinese Remainder Theorem gives a ring isomorphism

$$\Psi_d : \mathbb{Z}/d\mathbb{Z} \rightarrow \prod_i \mathbb{Z}/p_i^{r_i}\mathbb{Z}.$$

This is the *anchored CRT position* of x . It satisfies

$$\Psi_d(x + y) = \Psi_d(x) + \Psi_d(y), \quad \Psi_d(xy) = \Psi_d(x) \Psi_d(y), \quad \Psi_d(x^k) = \Psi_d(x)^k.$$

In this paper, Ψ_d is the finest finite coordinate used: it distinguishes every residue of $\mathbb{Z}/d\mathbb{Z}$ and therefore restores full deterministic transport. Coarser coordinates are useful only insofar as they retain enough information for a particular operation. The purpose of the later necessity theorem is to show that weaker profiles frequently fail.

4 Mirror signatures

For odd prime-power components p^r , define the quadratic mirror signature of a unit $u \in (\mathbb{Z}/p^r\mathbb{Z})^\times$ by

$$\chi_{p^r}(u) = \begin{cases} +1, & u \text{ is a quadratic residue mod } p^r, \\ -1, & u \text{ is a quadratic nonresidue mod } p^r. \end{cases}$$

For an odd quotient component $q_{\text{odd}} = \prod_i p_i^{r_i}$, define the mirror signature vector

$$\chi(u) = (\chi_{p_1^{r_1}}(u), \dots, \chi_{p_m^{r_m}}(u)).$$

On odd quotient components,

$$\chi(u^k) = \chi(u)^k.$$

Thus odd exponents preserve mirror sign, while even exponents send every unit mirror signature into the square cell.

For even prime-power components, no identical quadratic-sign law is asserted here. The 2-primary behavior is carried by the CRT prime-power coordinate and gcd-depth coordinate. Mirror signature is therefore an odd-component refinement, not a replacement for full CRT position.

5 Gcd-depth and saturated valuation

For each prime $p \mid d$, define the saturated depth

$$e_p(x) = \min(r_p, v_p(x)).$$

Then

$$g_d(x) = \prod_{p \mid d} p^{e_p(x)}.$$

For multiplication,

$$e_p(xy) = \min(r_p, e_p(x) + e_p(y)).$$

Equivalently, $g_d(xy)$ is determined componentwise by saturated addition of the prime-depths of $g_d(x)$ and $g_d(y)$.

This is the ring-native gcd-depth product law. It is stronger and cleaner than using raw representative valuations, because all depth is measured inside $\mathbb{Z}/d\mathbb{Z}$.

6 The row theorem

Theorem 1 (Row gcd-flow and quotient-unit closure). *Let $d \geq 2$, $x \in \mathbb{Z}/d\mathbb{Z}$, and consider the row $x, x^2, x^3, \dots$. For each $p \mid d$,*

$$e_p(x^k) = \min(r_p, k e_p(x)).$$

Therefore the gcd-depth sequence $g_d(x), g_d(x^2), \dots$ is monotone nondecreasing under divisibility and stabilizes after finitely many steps. Once the stable gcd-depth is reached, the row evolves inside a quotient-unit cycle.

Proof. For every prime $p \mid d$, $v_p(x^k) = k v_p(x)$. Saturating at r_p gives $e_p(x^k) = \min(r_p, k e_p(x))$. This is monotone in k and bounded by r_p , so it stabilizes.

Let g be the stabilized gcd-depth and write $x^k = gu_k$ with $q = d/g$. After stabilization, u_k is a unit modulo q . The row continues by multiplying by the surviving quotient image of x , so the post-stabilization row lies in the finite group $(\mathbb{Z}/q\mathbb{Z})^\times$. Every orbit in a finite group is periodic. $\square$

7 The column theorem

Theorem 2 (Column exponent action). *Fix $k \geq 1$. The k -th column of the table is the map*

$$C_k : \mathbb{Z}/d\mathbb{Z} \rightarrow \mathbb{Z}/d\mathbb{Z}, \quad C_k(x) = x^k.$$

For every x , $g_d(x) \mid g_d(x^k)$. On each prime component, $e_p(x^k) = \min(r_p, k e_p(x))$. On odd quotient-unit components, $\chi(u^k) = \chi(u)^k$.

Thus columns are exponent operators on gcd-depth, quotient-unit position, and odd-component mirror signatures.

Proof. The depth identity follows from $v_p(x^k) = k v_p(x)$ with saturation at r_p . Hence $e_p(x) \leq e_p(x^k)$, so $g_d(x) \mid g_d(x^k)$. For odd quotient units, quadratic character multiplicativity gives $\chi(u^k) = \chi(u)^k$. Thus the column map acts simultaneously on depth, quotient unit, and odd-component mirror signature. $\square$

8 Row-column ontology

Theorem 3 (Row-column ontology). *The table $T_d(x, k) = x^k \pmod{d}$ is a two-directional finite transport object. Rows fix x and vary k : they are governed by gcd-depth flow into quotient-unit cycling. Columns fix k and vary x : they are exponent operators acting globally on gcd strata, quotient units, and odd-component mirror signatures.*

Proof. The row statement is Theorem 1. The column statement is Theorem 2. Together they show that the table is not merely a list of powers; it is a two-directional projection of finite-ring transport. $\square$

9 Multiplicative transport theorem

Theorem 4 (Multiplicative stratum closure and quotient-unit cycling). *Let $d \geq 2$, $a, x \in \mathbb{Z}/d\mathbb{Z}$, and define the multiplicative transport path*

$$x_0 = x, \quad x_{n+1} = ax_n \pmod{d}.$$

Write $d = \prod_p p^{r_p}$, $e_p(n) = \min(r_p, v_p(x_n))$. Then:

1. *The gcd-depth vector evolves by saturated addition, $e_p(n+1) = \min(r_p, e_p(n) + v_p(a))$.*
2. *The sequence $e_p(n)$ is monotone nondecreasing and stabilizes after finitely many steps.*
3. *Once the stable gcd-depth is reached, write $g = \gcd(x_n, d)$, $q = d/g$, $x_n = gu_n$ with $u_n \in (\mathbb{Z}/q\mathbb{Z})^\times$. Then a is a unit modulo q , and the post-transient dynamics are $u_{n+1} \equiv au_n \pmod{q}$.*
4. *Every multiplicative path is eventually periodic. The post-transient cycle through u_n has length $\text{ord}_q(a)$, where q is the surviving quotient modulus.*

Proof. For each prime $p \mid d$, $v_p(ax_n) = v_p(a) + v_p(x_n)$. Since gcd-depths are measured inside $\mathbb{Z}/d\mathbb{Z}$, they saturate at r_p . Hence $e_p(n+1) = \min(r_p, e_p(n) + v_p(a))$. This proves the saturated valuation update. Monotonicity and finite stabilization follow because each $e_p(n)$ is a nondecreasing integer bounded above by r_p .

Let $g = \prod_p p^{e_p(\infty)}$ be the stable gcd-depth and $q = d/g$. For any prime $p \mid q$, the stable depth satisfies $e_p(\infty) < r_p$. If $v_p(a) > 0$, then multiplication by a would increase $e_p(n)$, contradicting stability. Therefore $v_p(a) = 0$ for every $p \mid q$. Hence $\gcd(a, q) = 1$. Thus a is a unit modulo q .

Since $x_n = gu_n$ and $\gcd(u_n, q) = 1$, $gu_{n+1} \equiv agu_n \pmod{d}$. Dividing by g inside the quotient gives $u_{n+1} \equiv au_n \pmod{q}$. Because a and u_n are units modulo q , the post-transient path lies in the finite group $(\mathbb{Z}/q\mathbb{Z})^\times$. The least positive ℓ satisfying $u_{n+\ell} = u_n$ satisfies $a^\ell u_n \equiv u_n \pmod{q}$. Multiplying by u_n^{-1} gives $a^\ell \equiv 1 \pmod{q}$. Thus the cycle has length $\ell = \text{ord}_q(a)$. $\square$

10 Additive transport theorem

Theorem 5 (Additive orbit closure). *Let $a \in \mathbb{Z}/d\mathbb{Z}$ and define $A_a(x) = x + a \pmod{d}$. Then A_a decomposes $\mathbb{Z}/d\mathbb{Z}$ into $\gcd(a, d)$ cycles, each of length $d/\gcd(a, d)$. In particular, A_a is one full cycle exactly when $\gcd(a, d) = 1$.*

Proof. The orbit of x under A_a is $x, x + a, x + 2a, \dots$. The least positive n for which $x + na \equiv x \pmod{d}$ is the least positive n satisfying $na \equiv 0 \pmod{d}$. This occurs at $n = d/\gcd(a, d)$. Therefore each cycle has length $d/\gcd(a, d)$. Since the set has d residues, the number of cycles is $\gcd(a, d)$. $\square$

11 Addition as anchored CRT transport

Theorem 6 (Addition as anchored CRT positional transport). *Let $d = \prod_i p_i^{r_i}$ and let $\Psi_d : \mathbb{Z}/d\mathbb{Z} \rightarrow \prod_i \mathbb{Z}/p_i^{r_i}\mathbb{Z}$ be the Chinese Remainder coordinate map. Then addition is determined componentwise by*

$$\Psi_d(x + y) = \Psi_d(x) + \Psi_d(y).$$

Equivalently, addition is deterministic once anchored CRT position is retained.

Proof. The componentwise identity is the standard CRT compatibility of addition, $(x+y) \bmod p_i^{r_i} = (x \bmod p_i^{r_i}) + (y \bmod p_i^{r_i}) \pmod{p_i^{r_i}}$. The full CRT position of x and y therefore determines the full CRT position of $x + y$, and hence determines $x + y \bmod d$. $\square$

Remark 7. Addition is generally *not* determined by gcd-depth, quotient modulus, quotient-unit order, or mirror signature alone. That insufficiency is proved explicitly in Theorem 10 by ambiguity witnesses. The role of Theorem 6 is the positive statement: anchored CRT position restores deterministic additive transport.

12 Projective compatibility

Theorem 8 (Cross-modulus projection compatibility). *Let $d \mid D$, and let $\pi_{D \rightarrow d} : \mathbb{Z}/D\mathbb{Z} \rightarrow \mathbb{Z}/d\mathbb{Z}$ be the natural projection. Then for all $x, y \in \mathbb{Z}/D\mathbb{Z}$ and all $k \geq 1$,*

$$\pi_{D \rightarrow d}(x + y) = \pi_{D \rightarrow d}(x) + \pi_{D \rightarrow d}(y), \quad \pi_{D \rightarrow d}(xy) = \pi_{D \rightarrow d}(x) \pi_{D \rightarrow d}(y),$$

and $\pi_{D \rightarrow d}(x^k) = \pi_{D \rightarrow d}(x)^k$. Therefore tile transport is compatible across moduli.

Proof. The map $\pi_{D \rightarrow d}$ is a ring homomorphism. Ring homomorphisms preserve addition and multiplication. Since powers are iterated multiplication, they preserve powers as well. $\square$

13 Transport-deterministic profiles

Definition 9 (Transport-deterministic profile). Let $S_d = \mathbb{Z}/d\mathbb{Z}$, and let $F : S_d^m \rightarrow S_d$ be a transport operation (addition, multiplication, or exponentiation). Let $\pi : S_d \rightarrow P$ be a profile map assigning to each residue a finite coordinate profile. Then π is *transport-deterministic for F* if, for all inputs $x_1, \dots, x_m$ and $y_1, \dots, y_m$,

$$\pi(x_i) = \pi(y_i) \text{ for all } i \implies \pi(F(x_1, \dots, x_m)) = \pi(F(y_1, \dots, y_m)).$$

If this implication fails, then π *loses transport information for F* . A coordinate is *transport-necessary* relative to a target profile if deleting that coordinate produces a profile that is no longer transport-deterministic.

14 Coordinate necessity theorem

Theorem 10 (Coordinate necessity). *Let the full transport coordinate of a residue $x \in \mathbb{Z}/d\mathbb{Z}$ include*

$$(g_d(x), q_d(x), u_d(x), \Psi_d(x), \chi(x)),$$

where $g_d(x) = \gcd(x, d)$, $q_d(x) = d/g_d(x)$, $u_d(x) = x/g_d(x) \pmod{q_d(x)}$, $\Psi_d(x)$ is the CRT prime-power position, and $\chi(x)$ is the odd-component mirror signature where defined. Then:

1. *Gcd-depth alone does not determine addition.*
2. *Gcd-depth plus quotient modulus does not determine addition.*
3. *Gcd-depth, quotient modulus, and mirror signature do not determine addition.*
4. *Quotient-unit data alone does not determine multiplication.*
5. *Mirror-signature data alone does not determine multiplication.*
6. *Full quotient-unit position determines row, column, multiplication, and power transport on each quotient once gcd-depth is accounted for.*
7. *Full CRT prime-power position determines addition, multiplication, and power transport.*

Proof. We give explicit witnesses. Each witness consists of two or more input pairs with the same reduced profile but different output profiles.

1. Gcd-depth alone does not determine addition. Take $d = 9$. The residues 1, 2, 4, 5, 7, 8 all have gcd-depth 1. Compare $1 + 1 \equiv 2 \pmod{9}$ and $1 + 2 \equiv 3 \pmod{9}$. The two input pairs have the same gcd-depth profile $((1, 1), (1, 1))$, but $\gcd(2, 9) = 1$ while $\gcd(3, 9) = 3$.

2. Gcd-depth plus quotient modulus does not determine addition. Again take $d = 9$. For units 1 and 2, $g_d(1) = g_d(2) = 1$ and $q_d(1) = q_d(2) = 9$. The pairs (1, 1) and (1, 2) have the same (g, q) -profile, but $1 + 1 \equiv 2$ has output $(g, q) = (1, 9)$, while $1 + 2 \equiv 3$ has output $(g, q) = (3, 3)$.

3. (g, q, χ) does not determine addition. Take $d = 9$. In $(\mathbb{Z}/9\mathbb{Z})^\times$, the squares are $\{1, 4, 7\}$ and the nonsquares are $\{2, 5, 8\}$. Consider the input profile $((1, 9, +), (1, 9, -))$. The pairs (1, 2), (1, 5), (1, 8) share that profile, but $1 + 2 = 3$, $1 + 5 = 6$, $1 + 8 \equiv 0 \pmod{9}$. The residues 3 and 6 share $(g, q) = (3, 3)$ but differ in reduced quotient unit: $u_d(3) = 1 \pmod{3}$, $u_d(6) = 2 \pmod{3}$. The residue 0 has terminal profile $(g, q) = (9, 1)$. Anchored CRT position is required.

4. Quotient-unit data alone does not determine multiplication. Take $d = 9$. The residues 1 and 3 both have reduced quotient representative $u = 1$: $1 = 1 \cdot 1$, $g = 1, q = 9$; $3 = 3 \cdot 1$, $g = 3, q = 3$. Compare multiplication pairs $(1, 1)$ and $(3, 3)$: same u -only input profile $(1, 1)$, but $1 \cdot 1 \equiv 1 \pmod{9}$ and $3 \cdot 3 \equiv 0 \pmod{9}$. Different transport profiles.

5. Mirror signature alone does not determine multiplication. Take $d = 9$. Mirror signature records square-class information on odd quotient units, but it does not remember gcd-depth or whether a residue has already entered a nonunit stratum. Compare $1 \cdot 1 \equiv 1 \pmod{9}$ with $3 \cdot 3 \equiv 0 \pmod{9}$.

6. Quotient-unit position determines quotient transport. Once g and $q = d/g$ are fixed and $x = gu$ with $u \in (\mathbb{Z}/q\mathbb{Z})^\times$, multiplication inside the surviving quotient is unit transport, $u \mapsto au \pmod{q}$, provided a is a unit modulo q . Theorem 4 proves that after gcd-depth stabilizes, the surviving multiplier is necessarily a unit modulo q . Similarly, exponentiation is determined on the quotient by $u \mapsto u^k \pmod{q}$.

7. CRT prime-power position determines full transport. The CRT map $\Psi_d : \mathbb{Z}/d\mathbb{Z} \rightarrow \prod_{p^r \parallel d} \mathbb{Z}/p^r\mathbb{Z}$ is a ring isomorphism, so $\Psi_d(x + y) = \Psi_d(x) + \Psi_d(y)$, $\Psi_d(xy) = \Psi_d(x)\Psi_d(y)$, $\Psi_d(x^k) = \Psi_d(x)^k$. CRT prime-power position therefore determines addition, multiplication, and power transport exactly. $\square$

Remark 11 (Scope). The theorem proves necessity relative to the transport-coordinate family introduced here. It does not claim that no mathematically equivalent coordinate system could encode the same information. The precise statement is: if the coordinates are weakened by deleting gcd-depth, quotient modulus, quotient-unit position, mirror refinement, or anchored CRT position, deterministic transport fails on the induced profile classes.

15 Full transport closure theorem

Theorem 12 (Full transport closure). *Let $d \geq 2$ and $T_d(x, k) = x^k \pmod{d}$. The following transport laws close over finite tile coordinates:*

1. Rows $k \mapsto x^k$ evolve by saturated gcd-depth flow and eventually enter quotient-unit cycles.
2. Columns $x \mapsto x^k$ act as exponent operators on gcd-depth, quotient-unit position, and odd-component mirror signatures.
3. Addition $x + y$ is deterministic in anchored CRT prime-power position.
4. Multiplication xy is determined by saturated gcd-depth folding and quotient-unit multiplication.
5. Multiplicative transport paths are finite functional graphs whose components are transient-to-cycle paths.
6. Post-transient multiplicative cycles have length $\text{ord}_q(a)$, where q is the surviving quotient modulus and a is the multiplier.
7. Cross-modulus projection $D \rightarrow d$ preserves addition, multiplication, and powers whenever $d \mid D$.

Proof. Rows are handled by Theorem 1. Columns are handled by Theorem 2. Addition closes by the additive orbit theorem and is deterministically represented by anchored CRT position by Theorem 6. Multiplication closes by Theorem 4, with monotone stabilizing gcd-depth dynamics

and post-stabilization quotient-unit cycles of length $\text{ord}_q(a)$. Projection compatibility is the cross-modulus theorem.

Since $\mathbb{Z}/d\mathbb{Z}$ is finite, every deterministic self-map decomposes into finite functional graph components, each consisting of a directed cycle with rooted transient trees feeding into it. The previous theorems identify the algebraic structure of those components for rows, columns, addition, multiplication, and projections. $\square$

16 Audit grid

The certificate computations are not used as substitutes for proof. The proofs above are algebraic. The certificates serve three narrower purposes: to verify implementation consistency; to expose ambiguity witnesses for deleted-coordinate profiles; and to test the theorems across representative modulus types.

The modulus list was chosen to cover small primes (2, 3, 5, 7, 73); prime powers (4, 8, 9, 27); squarefree composites (6, 15, 21, 30, 35, 77, 105, 210); nonsquarefree composites (12, 18, 24, 45, 63, 90); and a larger semiprime ($1081 = 23 \cdot 47$). This selection is not a claim of proof by enumeration; it is an audit grid designed to check the algebra across prime, prime-power, squarefree, nonsquarefree, and semiprime cases.

The primary certificate gates are: row gcd-flow monotonicity; row eventual cycle closure; column gcd-depth monotonicity; odd mirror exponent law; saturated gcd-depth product law; additive cycle law; additive anchored CRT closure; multiplicative functional graph closure; multiplicative transient depth law; multiplicative quotient-order law. The full transport closure certificate passed all final exact-law gates after the saturated gcd-depth law was stated in ring-native form.

The coordinate necessity certificate then tested whether deleting coordinates causes induced profile classes to lose deterministic transport. The resulting pattern was: gcd-depth alone is insufficient for addition; quotient modulus alone is insufficient for addition; mirror signature alone is insufficient for multiplication; (g, q, χ) helps but does not determine addition; quotient-unit position restores quotient transport; CRT prime-power position restores full addition, multiplication, and power transport.

17 Conclusion

The finite power table $T_d(x, k) = x^k \pmod{d}$ has a disciplined row-column and additive-multiplicative transport ontology. The coordinate stack

$$(g_d(x), q_d(x), u_d(x), \Psi_d(x), \chi(x))$$

is not chosen because it is visually convenient. It is forced by the demand that the finite transport laws close deterministically.

Rows flow. Columns act. Addition walks anchored CRT position. Multiplication folds through gcd-depth and then cycles by quotient-unit order. Cross-modulus projection preserves the operations.

The result is intentionally conservative. It is a theorem about finite rings and their transport coordinates.

References

- [1] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, 1976.

- [2] W.-S. Chou and I. E. Shparlinski, *On the cycle structure of repeated exponentiation modulo a prime*, J. Number Theory **107** (2004), 345–356.
- [3] J. Grieshop, *Explicit constants for Fejér-type smoothing on finite cyclic groups*, arXiv:2512.03085 [math.CA], 2025.
- [4] J. Grieshop, *On the failure of step-response tests to certify admissibility of spectral averaging operators*, arXiv:2602.20185 [math.CA], 2026.
- [5] J. Grieshop, *Marithmetics (archival release)*, Zenodo, 2026. doi:10.5281/zenodo.19424837.
- [6] J. Grieshop, *Marithmetics: source repository, scripts, and demonstrations*, GitHub repository, 2026. <https://github.com/public-arch/Marithmetics>.
- [7] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.
- [8] J. Grieshop, *Power-residue tiles as finite arithmetic objects*, companion paper, 2026.
- [9] T. D. Rogers, *The graph of the square mapping on the prime fields*, Discrete Math. **148** (1996), 317–324.
- [10] L. DeVille, *The shape of $x^2 \bmod n$* , arXiv:2207.10512 [math.CO], 2022.
- [11] T. Vasiga and J. Shallit, *On the iteration of certain quadratic maps over $\text{GF}(p)$* , Discrete Math. **277** (2004), 219–240.